Assignment 3

1.

```
drop table buyer;
drop table developer;
drop table house;
drop table insurance;

create table buyer(
 buyerid number,
 fname varchar2(20),
 lname varchar2(20),
 ic number,
 address varchar2(30),
 phone varchar2(15),
 constraint pk_buyer primary key (buyerid)
);

create table developer(
 developerid number,
 name varchar2(20),
 address varchar2(50),
 constraint pk_developer primary key (developerid)
);

create table house(
 houseno number,
 buyerid number not null,
 developerid number not null,
 location varchar2(50),
 type varchar2(15),
 price number(12,2),
 constraint pk_house primary key (houseno),
```

```
 constraint fk_house_buyer foreign key(buyerid) references buyer(buyerid),
 constraint fk_house_developer foreign key (developerid) references developer(d
);

create table insurance(
 policy number,
 houseno number,
 duration varchar2(15),
 amount number(12,2),
 constraint pk_insurance primary key(policy),
 constraint fk_insurance_house foreign key(houseno) references house(houseno
);

insert into buyer values (
  899, 'Qaseh','Qassim',771103101234,'Bandar Baru Bangi','019-1234567'
);
insert into buyer values (
  123, 'Khalid', 'Ahmad', 671103101443, 'Seksyen 12, Shah Alam', '012-3451234'
);
insert into buyer values (
  234, 'Halimah', 'Hamdan', 870113041674, 'Jelebu, Seremban', '013-1209876'
);
insert into buyer values (
  789, 'Zainal', 'Abidin', 770723011337, 'Kg Dato Keramat', null
);
insert into buyer values (
  666, 'Kalsum', 'Hassan', 741014141844, 'Equine, Serdang', '010-2332244'
);
insert into buyer values (
  107, 'Zamani', 'Zakaria', 800303031339, 'Jalan Reko, Kajang', '019-4561239'
);
insert into developer values (
  10, 'Maju Jaya', 'Lot 15, Kajang, Selangor'
);
insert into developer values (
  11, 'Exploria', 'PKNS, Bangi, Selangor'
```

```
);
insert into developer values (
  13, 'Kemuning', '123, Sungai Buluh, Selangor'
);
insert into developer values (
  15, 'Temasik', 'Blok 156, Rawang, Selangor'
);
insert into house values (
  1000, 123, 15, 'Seksyen 9, Bangi, Selangor', 'Terrace', 450000
);
insert into house values (
  1050, 234, 15, 'Sungai Dua, Nilai, N Sembilan', 'Terrace', 380000
);
insert into house values (
  1020, 666, 13, 'Equine, Seri Kembangan, Selangor', 'Apartment', 285000
);
insert into house values (
  1080, 234, 11, 'Bandar Mutiara, Bukit Katil, Melaka', 'Semi-D', 600000
);
insert into house values (
  1040, 789, 13, 'Bukit Mahkota, Seremban, Selangor', 'Semi-D', 899000
);
insert into house values (
  1030, 107, 15, 'Puncak Alam, Shah Alam, Selangor', 'Apartment', 130000
);
insert into insurance values (
  1987654, 1020, '5 years', 500000
);
insert into insurance values (
  1234567, 1040, '3 years', 935000
);
insert into insurance values (
  1336699, 1080, '1 year', 123000
);
insert into insurance values (
```

```
  1155583, 1000, '1 year', 100000
);
```

2.

```
select * from buyer;
select * from developer;
select * from house;
select * from insurance;
```

3.

```
alter table developer modify(address varchar2(50));
```

4.

```
select buyerid, fname ||' ' || lname as "Name", NVL(phone, '99999') as phone
from buyer;
```

5.

```
select type from house
order by type desc;
```

6.

```
select houseno, price from house
where price < (select avg(price) from house)
```

7.

```
select b.buyerid, count(h.houseno) as Total
from buyer b
left join house h
```

```
on b.buyerid = h.buyerid
group by b.buyerid
```

8.

```
select d.name, h.type, h.price
from developer d
join house h
on d.developerid = h.developerid
where h.location like '%Selangor' and h.price between 250000 and 500000
```

9.

```
select houseno, price,
case
when price <= 250000 then 'Kos Rendah'
when price <= 500000 then 'Mampu Milik'
else 'Komuniti'
end as category
from house
```

10.

```
select buyerid, sum(price) as totalprice
from house
group by buyerid
having count(*) > 1
```

11.

```
update insurance i
set i.amount = i.amount * 1.1
where i.houseno in (select h.houseno from house h where h.type = 'Semi-D')
```

12.

```
delete from developer where developerid not in (select developerid from house)
```